

COST MINIMIZATION PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

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PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Problem 1

Poizy the gardener is an avid gardening enthusiast. He has found that the number of happy plants, h , depends on the amount of light l , and water, w . In fact, Poizy noticed that plants require twice as much light as water, and any more or less is wasted. Thus, Poizy's production function is $h = \min\{l, 2w\}$.

- (a) Suppose that Poizy is using 1 unit of light. What is the least amount of water he can use and still produce a happy plant?
- (b) Suppose that Poizy wants to produce 4 happy plants. What are the minimum amounts of light and water required?
- (c) What is Poizy's conditional input demand function for light? What is Poizy's conditional input demand function for water?
- (d) If each unit of light costs w_1 and each unit of water costs w_2 , Poizy's cost function $C(w_1, w_2, h)$ is?

Problem 2

Poizy, from the previous task, is dealing with an acquaintance, Chatterbox Chucklehead (CC), who happens to be also an avid gardener and, more importantly, someone who never misses a chance to talk. CC claims to have discovered an alternative method of gardening: according to him, happy plants only require fertilizer¹ and talk. In fact, he insists that these two inputs are perfect substitutes! Thus, suppose that f is the number of bags of fertilizer used and t is the number of hours CC talks to the plants; then the number of happy plants produced is exactly $h = t + 2f$. Suppose fertilizer costs w_f per bag and talk costs w_t per hour.

- (a) If CC uses no fertilizer, how many hours of talk must be devoted if CC wants one happy plant?
- (b) If CC doesn't talk to the plants at all, how many bags of fertilizer will be needed for one happy plant?
- (c) If $w_t < w_f/2$, would it be cheaper for CC to use fertilizer or talk to raise one happy plant?
- (d) What is CC's cost function $c(w_f, w_t, h)$?
- (e) What is CC's conditional input demand for talk, $t(w_f, w_t, h)$?

¹Please recall where fertilizer comes from.

Problem 3

NU's cafeteria produces meals, using two inputs, "gas burners" g , and a mystery ingredient x . I am not at liberty to say what the mystery ingredient is, but I have on good authority that "meat and fungus are involved." The cafeteria's production function is $f(x, g) = gx^2$, where x is the amount of the mystery ingredient, g are gas burners, and $f(x, g)$ is the amount of meals produced.

- (a) Does the cafeteria have increasing, constant, or decreasing returns to scale.
- (b) For the rest of the problem, suppose that we are considering the short-run, so that the amount of gas burners is fixed at $g_0 = 10$. How many units of the mystery ingredient x takes to produce 1440 meals? If the input costs are w_x per unit of x and w_g for running each burner, what is the short-run cost to produce 1440 meals?
- (c) How many units of x are needed to produce y meals? If input costs are w_x and w_g , what is the cost to produce y means?
- (d) Provide expressions for the average total cost, average variable cost, average fixed cost, and marginal cost curves as functions of y , w_x and w_g .